



Divid: Disentangled Spatial-Temporal Modeling within LLMs for Temporally Grounded Video Understanding

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❖ Introduction

Motivation:

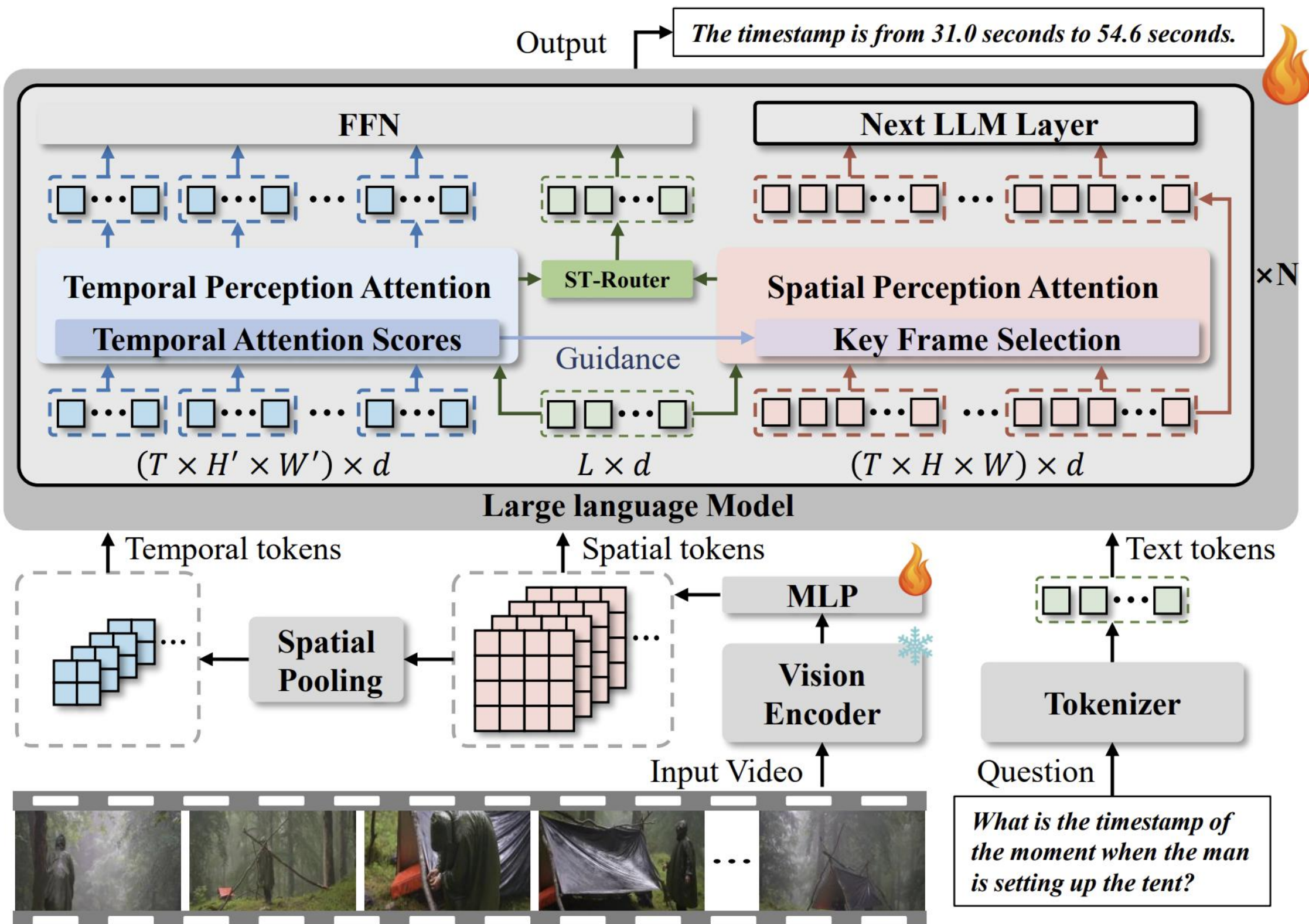
Although Video LLMs have advanced video understanding, temporally grounded understanding in long-form videos remains challenging. Existing methods often flatten frames into long token sequences, leading to high computation cost, context bottlenecks, and limited temporal precision. Moreover, spatial and temporal reasoning is still entangled inside the LLM, while query-agnostic frame selection may miss critical details. Existing timestamp-supervised datasets also suffer from a trade-off between scale and annotation quality, limiting fine-grained temporal grounding.

Our Contribution:

- We propose Divid to separately model temporal dynamics and spatial details inside the LLM decoder.
- We design query-aware keyframe selection and a soft-router for adaptive spatial-temporal fusion.
- We construct TempGCap with 559K timestamp-grounded samples and demonstrate superior accuracy and efficiency on multiple benchmarks.

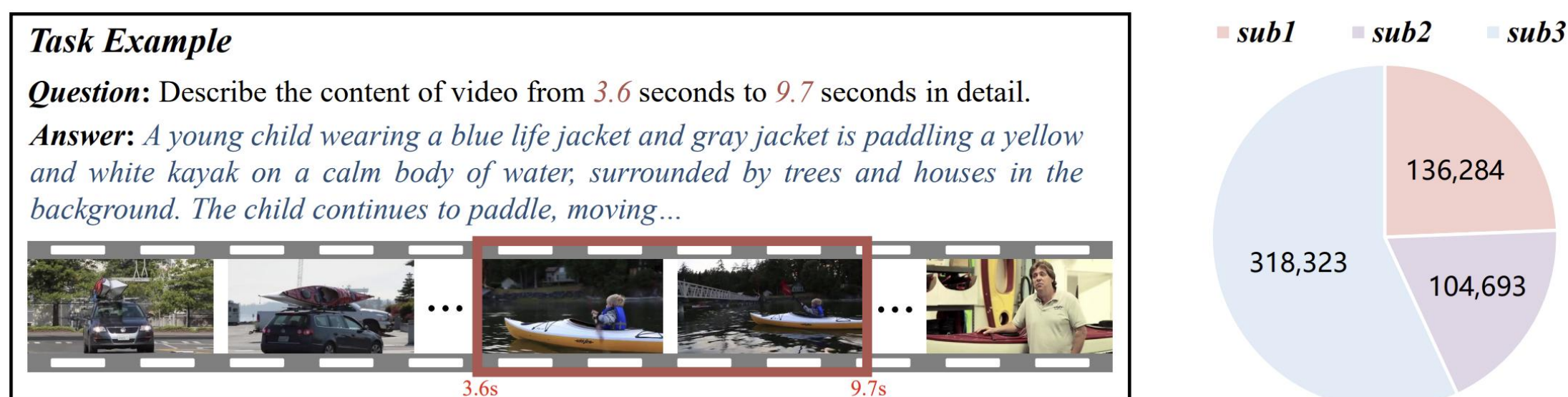
❖ Divid Framework

- **Temporal Branch:** models densely sampled low-resolution frames to capture global temporal dynamics.
- **Spatial Branch:** selects query-relevant high-resolution keyframes for detailed spatial understanding.
- **ST-Router:** adaptively fuses temporal and spatial features at the token level for accurate grounding and efficient reasoning.



❖ TempGCap Dataset

TempGCap is a large-scale dataset for timestamp-guided captioning with 559K video-text pairs from 428K videos. **Sub1** enriches manually annotated untrimmed videos with segment-level captions. **Sub2** restores long-video context from captioned clips and refines timestamps. **Sub3** creates pseudo-untrimmed videos by stitching short clips together. These three sources jointly provide precise temporal supervision and diverse video coverage for temporally grounded video understanding.



❖ Evaluation Results

Charades-STA					CGBench				
Method	Size	R@0.5	R@0.7	mIoU	Method	Size	mIoU	R@IoU	A@IoU
(FT) Moment-DETR (Lei et al., 2021)	–	52.1	30.6	45.5	GPT-4o (OpenAI, 2024a)	–	5.62	8.30	4.38
(FT) UniVTG (Lin et al., 2023c)	–	58.1	35.6	50.1	GPT-4o-mini (OpenAI, 2024b)	–	3.75	5.18	2.21
(FT) R ² -Tuning (Liu et al., 2024e)	–	59.8	37.0	50.9	Gemini-1.5-Pro (Team, 2024a)	–	3.95	5.81	2.53
GPT-4o (OpenAI, 2024a)	–	–	–	35.7	Gemini-1.5-Flash (Team, 2024a)	–	3.67	5.44	2.45
Grounded-VideoLLM (Wang et al., 2024b)	4B	36.4	19.7	36.8	Claude-3.5-Sonnet	–	3.99	5.67	2.79
E.T. Chat (Liu et al., 2024f)	4B	45.9	20.0	42.3	Qwen2-VL (Wang et al., 2024d)	72B	3.58	5.32	2.54
Qwen2.5-VL (Bai et al., 2025)	3B	–	–	38.8	VITA (Zhang et al., 2024c)	8×7B	3.06	3.53	2.33
VTimeLLM (Huang et al., 2024a)	13B	34.3	14.7	34.6	ShareGPT4Video (Chen et al., 2024c)	16B	1.85	2.65	1.01
Qwen2.5-VL (Bai et al., 2025)	72B	–	–	50.9	Video-CCAM (Team, 2024b)	14B	2.63	3.53	1.76
Qwen2.5-VL (Bai et al., 2025)	7B	–	–	43.6	Chat-UniVi-v1.5 (Jin et al., 2023)	13B	2.07	2.53	1.20
TimeChat (Ren et al., 2024)	7B	32.2	13.4	–	LLaVA-OV (Liu et al., 2024g)	13B	1.63	1.78	1.01
Momentor (Qian et al., 2024a)	7B	26.6	11.8	28.5	Video-LLaVA (Lin et al., 2023a)	7B	1.13	1.96	0.59
HawkEye (Wang et al., 2024e)	7B	31.4	14.5	34.7	Video-LLaMA (Zhang et al., 2023c)	7B	1.21	1.87	0.84
ChatVTG (Qu et al., 2024)	7B	33.0	15.9	34.9	Videochat2 (Li et al., 2024a)	7B	1.28	1.98	0.94
VideoChat-TPO (Yan et al., 2024)	7B	40.2	20.8	38.1	Qwen-VL-Chat (Wang et al., 2023)	7B	0.89	1.19	0.42
VideoChat-T (Zeng et al., 2024)	7B	48.7	24.0	–	ST-LLM (Liu et al., 2024d)	7B	2.23	2.86	1.13
VideoChat-Flash (Li et al., 2024b)	7B	53.1	27.6	–	ViLA (Lin et al., 2023b)	8B	1.56	2.89	1.35
TimeMarker (Chen et al., 2024d)	8B	51.9	26.9	48.4	MiniCPM-v2.6 (OpenBMB, 2024)	8B	2.35	2.96	1.35
TimeSearch (Pan et al., 2025)	7B	52.4	24.5	48.6	LongVA (Zhang et al., 2024a)	7B	2.94	3.86	1.78
VideoMind (Liu et al., 2025)	1.5B	51.1	26.0	45.2	Kangaroo (Liu et al., 2024c)	8B	2.26	3.24	1.23
VideoMind (Liu et al., 2025)	7B	59.1	31.2	50.2	InternVL2 (Chen et al., 2024e)	7B	3.91	5.05	2.64
Divid (Ours)	1.5B	51.4	26.9	47.3	Divid (Ours)	1.5B	4.77	6.40	2.37
Divid (Ours)	7B	59.5	31.3	51.3	Divid (Ours)	7B	5.74	8.36	4.11

Next-GQA

Method	LLM Size	IoU			IoP			Acc@GQA
		R@0.3	R@0.5	mIoU	R@0.3	R@0.5	mIoP	
FrozenBiLM NG+ (Yang et al., 2022)	890M	13.5	6.1	9.6	28.5	23.7	24.2	17.5
VIOLETV2 (Fu et al., 2023)	–	4.3	1.3	–	–	25.1	23.3	12.8
SeViLA (Yu et al., 2023)	4B	29.2	13.8	21.7	–	34.7	22.9	16.6
LangRepo (Kahatapitiya et al., 2024)	8×7B	–	12.2	18.5	–	20.8	27.1	14.9
VideoStreaming (Qian et al., 2024b)	8.3B	–	13.3	19.4	–	22.0	30.8	18.1
LLoVi (Zhang et al., 2023a)	1.8T	15.3	20.0	–	–	36.6	37.3	24.3
HawkEye (Wang et al., 2024e)	7B	37.0	19.5	25.7	44.7	31.6	33.4	23.5
Grounded-VideoLLM (Wang et al., 2024b)	4B	30.2	18.0	21.1	42.6	34.4	34.5	26.7
VideoChat-TPO (Yan et al., 2024)	7B	41.2	23.4	27.7	47.5	32.8	35.6	25.5
VideoMind (Liu et al., 2025)	1.5B	45.2	23.2	28.6	51.3	30.6	35.6	25.2
VideoMind (Liu et al., 2025)	7B	50.2	25.8	31.4	56.0	35.3	39.0	28.2
Divid (Ours)	1.5B	47.5	26.8	32.9	52.0	33.7	38.9	26.4
Divid (Ours)	7B	51.3	27.5	34.5	54.6	38.2	40.8	29.2

ReXTime

Method	LLM Size	FT	R@0.3	R@0.5	mIoU	Acc@IoU
VTimeLLM (Huang et al., 2024a)	7B	✓	43.69	26.13	29.92	17.13
TimeChat (Ren et al., 2024)	7B	✓	40.13	21.42	26.29	10.92
VTimeLLM (Huang et al., 2024a)	7B	–	28.84	17.41	20.14	–
TimeChat (Ren et al., 2024)	7B	–	14.42	7.61	11.65	–
LITA (Huang et al., 2024b)	13B	–	29.49	16.29	21.49	–
VideoMind (Liu et al., 2025)	1.5B	–	34.31	22.69	24.83	17.26
VideoMind (Liu et al., 2025)	7B	–	38.22	25.52	27.61	20.20
Divid (Ours)	1.5B	–	40.50	26.82	29.71	18.57
Divid (Ours)	7B	–	42.56	31.05	35.78	22.26

❖ Analysis

Different Frameworks

Method	LLM TFLOPs	Charades-STA		ReXTime	
		R1@0.5	mIoU	mIoU	Acc@IoU
Full	28.2	51.55	47.76	30.16	19.11
Slow-Fast	16.2	50.65	46.71	29.49	18.45
Divid	10.5	51.37	47.33	29.71	18.57

Temporal Guidance

Method	Charades-STA		ReXTime	
	R1@0.5	mIoU	mIoU	Acc@IoU
Uniform	49.70	46.27	27.61	17.32
Weighted	48.62	45.43	26.44	16.78
Segment	50.96	47.20	28.82	17.59
Importance	50.86	47.10	28.83	17.92
Top-K	51.37	47.33	29.71	18.57

ST-Router

Method	Charades-STA		ReXTime	
	R1@0.5	mIoU	mIoU	Acc@IoU
Add	50.66	46.81	29.11	18.23
Weight	50.98	47.13	29.31	18.42
Soft-Router	51.37	47.33	29.71	18.57

Different Data

Method	Charades-STA		ReXTime	
	R1@0.5	mIoU	mIoU	Acc@IoU
Momentor-1M (Qian et al., 2024a)	48.96	44.90	26.77	15.43
Momentor (Qian et al., 2024a)	50.47	46.42	28.75	16.78
TempGCap	51.37	47.33	29.71	18.57

Ablation results confirm that Divid achieves an effective balance between efficiency and accuracy. **Top-K temporal guidance**, **Soft-Router fusion**, and **TempGCap** training data all consistently improve performance, demonstrating that query-aware keyframe selection, adaptive spatial-temporal fusion, and high-quality temporal supervision are critical for temporally grounded video understanding.



Qualitative visualization of Divid shows that the temporal branch guides high-resolution keyframe selection, helping the spatial branch capture fine-grained visual details and predict a segment close to the ground truth.